

Improved Functionality of GRNsight 6.0: a Web Application for Visualizing Gene Network Models

Ahmad R. Mersaghian*, Sarron A. Tadesse **, Ngan N. Tran**, Kam D. Dahlquist*, John David N. Dionisio**

*Department of Biology, **Department of Computer Science, Loyola Marymount University, 1 LMU Drive, Los Angeles, CA 90045

<http://dondi.github.io/GRNsight/>



GRNsight Automatically Lays Out Unweighted and Weighted Gene Regulatory Network Graphs

1. Network

- Demo files are provided.
- Can import and export Excel, SIF, or GraphML files.
- Can construct a network from SGD regulation data stored in a backend database.

2. Layout

- Users can select a force graph layout or grid layout.
- Force graph parameter sliders:
 - Link distance determines the minimum distance between nodes.
 - Nodes have a charge which repels or attracts other nodes.

3. Node

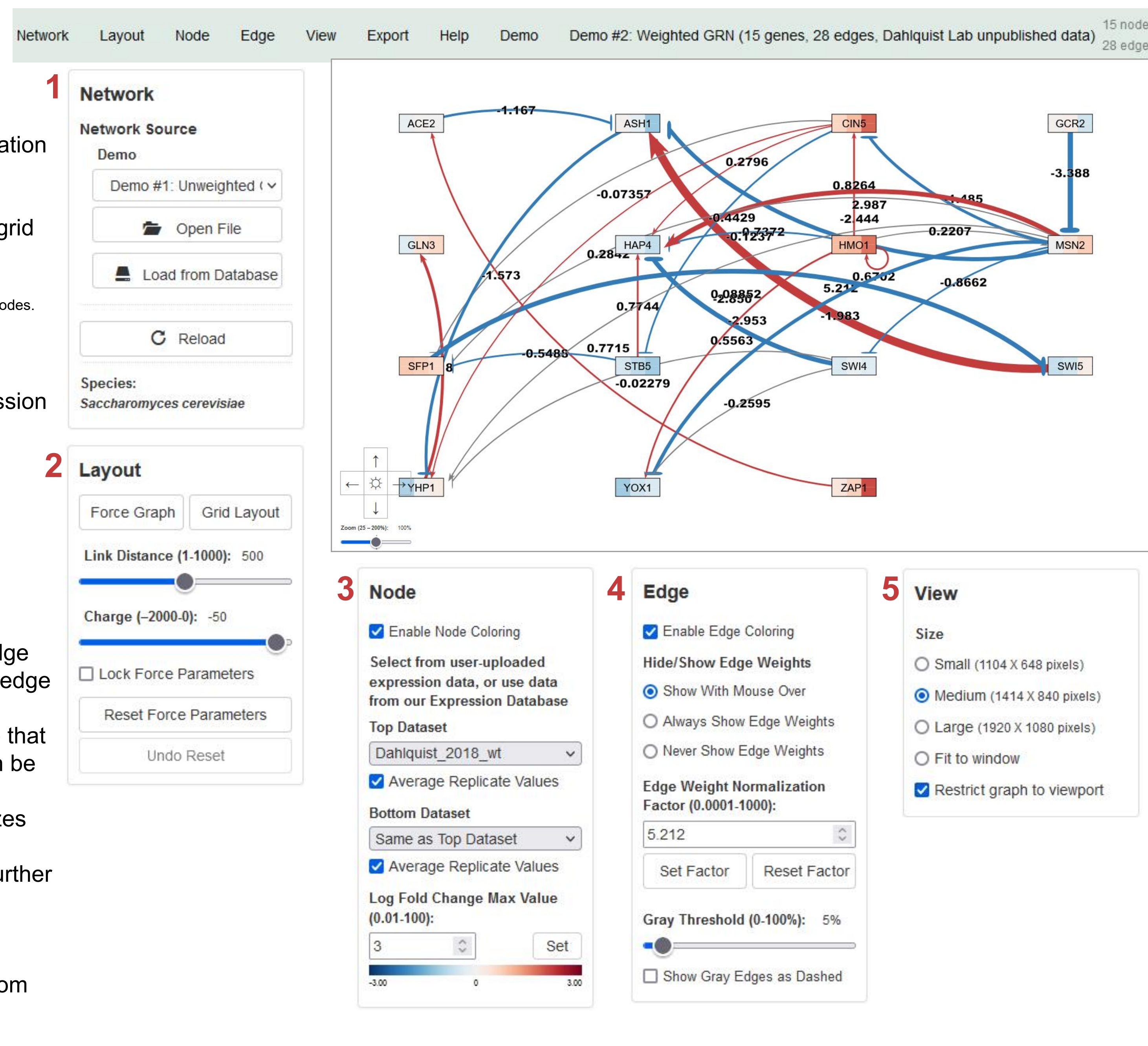
- Coloring can be based on user-uploaded expression data or from published expression datasets stored in a backend database.
- Datasets available:
 - Apweiler et al. 2012, GSE33098
 - Barreto et al. 2012, GSE24712
 - Dahlquist et al. 2018, GSE83656
 - Kitagawa et al. 2002, GSE9336
 - Thorsen et al. 2007, GSE6068

4. Edge

- Edge coloring can be toggled on and off.
- Buttons enable the user to always see edge weights, never see edge weights, or see edge weights upon mouseover.
- Allows user to set normalization factor so that edge thicknesses for different graphs can be rendered on the same scale.
- Changing the gray threshold deemphasizes weaker edges in the visualization.
- Presenting gray edges as dashed lines further deemphasizes weak edges.

5. View

- Multiple viewport sizes available.
- Graph bounding box can be separated from viewport.
- Zooming and scrolling available.



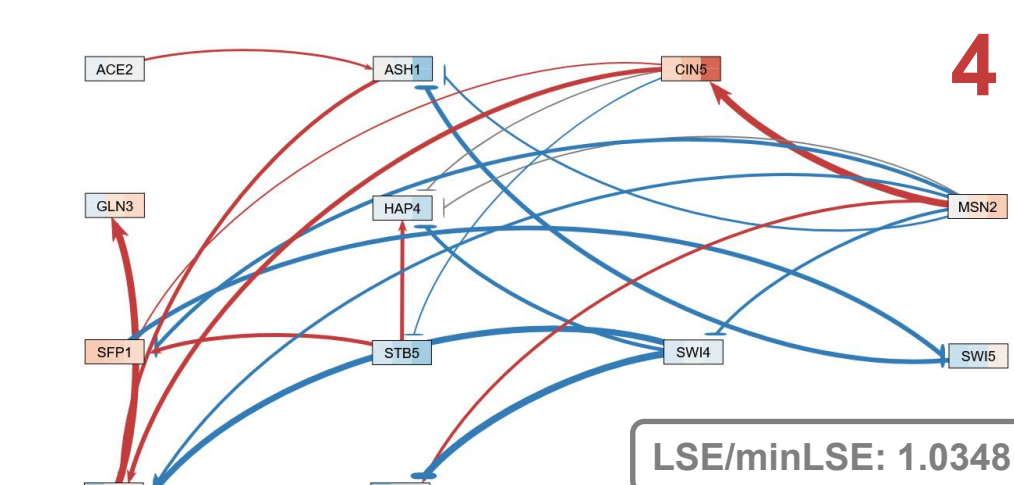
3. Dynamical Systems Modeling

- The exported Excel file is GRNmap compatible and can be used for further modeling.

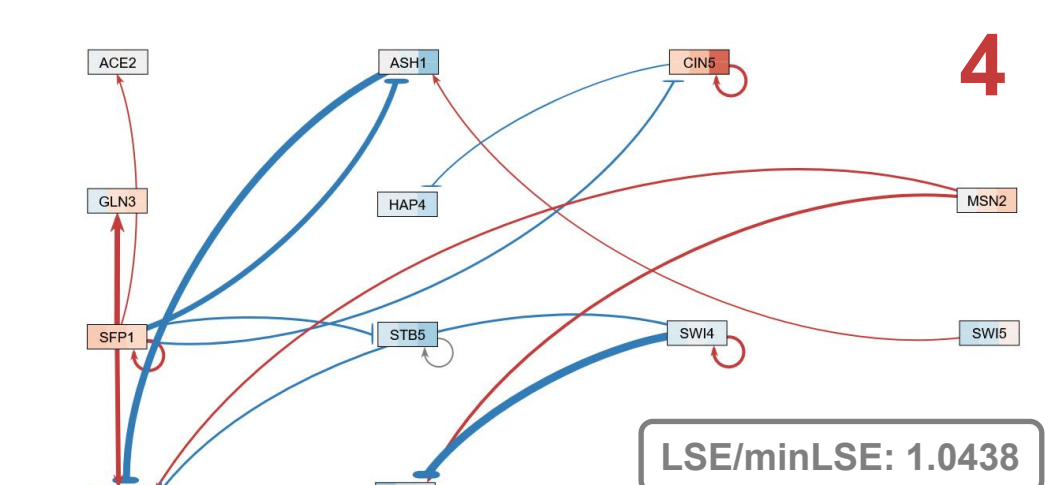
4. Visualization of Modeling Results

- The new model can be visualized in GRNsight.
- Workflow facilitates modeling multiple networks with multiple expression datasets more quickly.

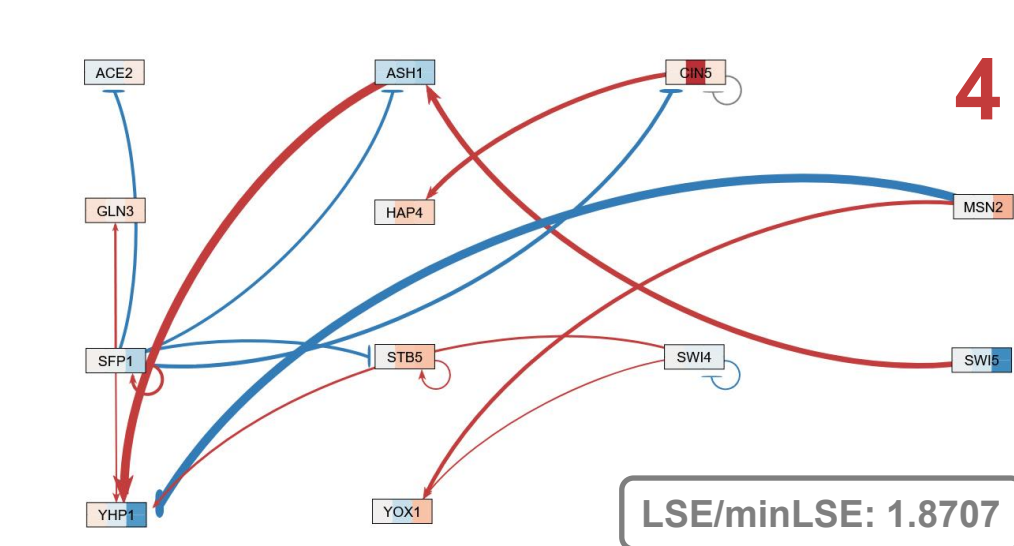
$$\frac{dx_i(t)}{dt} = \frac{P_i}{1 + \exp\left(-\sum_j (w_{ij}x_j(t) - b_i)\right)} - d_i x_i(t)$$



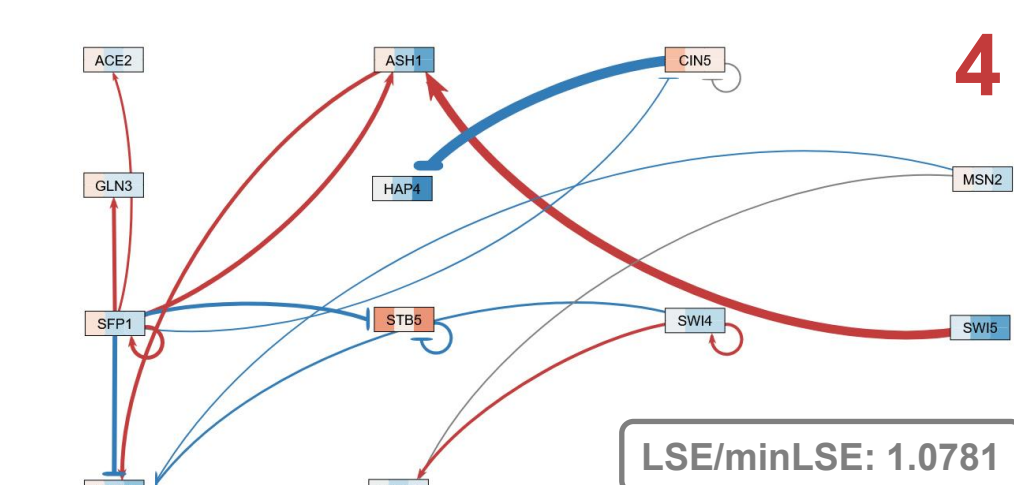
- 12-gene, 21 edge network from YEASTRACT
- Modeled with Dahlquist et al. (2018) expression data
- wt yeast subjected to 15, 30, and 60 minutes of cold shock at 13°C



- 12-gene, 17 edge network from SGD
- Modeled with Dahlquist et al. (2018) expression data
- wt yeast subjected to 15, 30, and 60 minutes of cold shock at 13°C



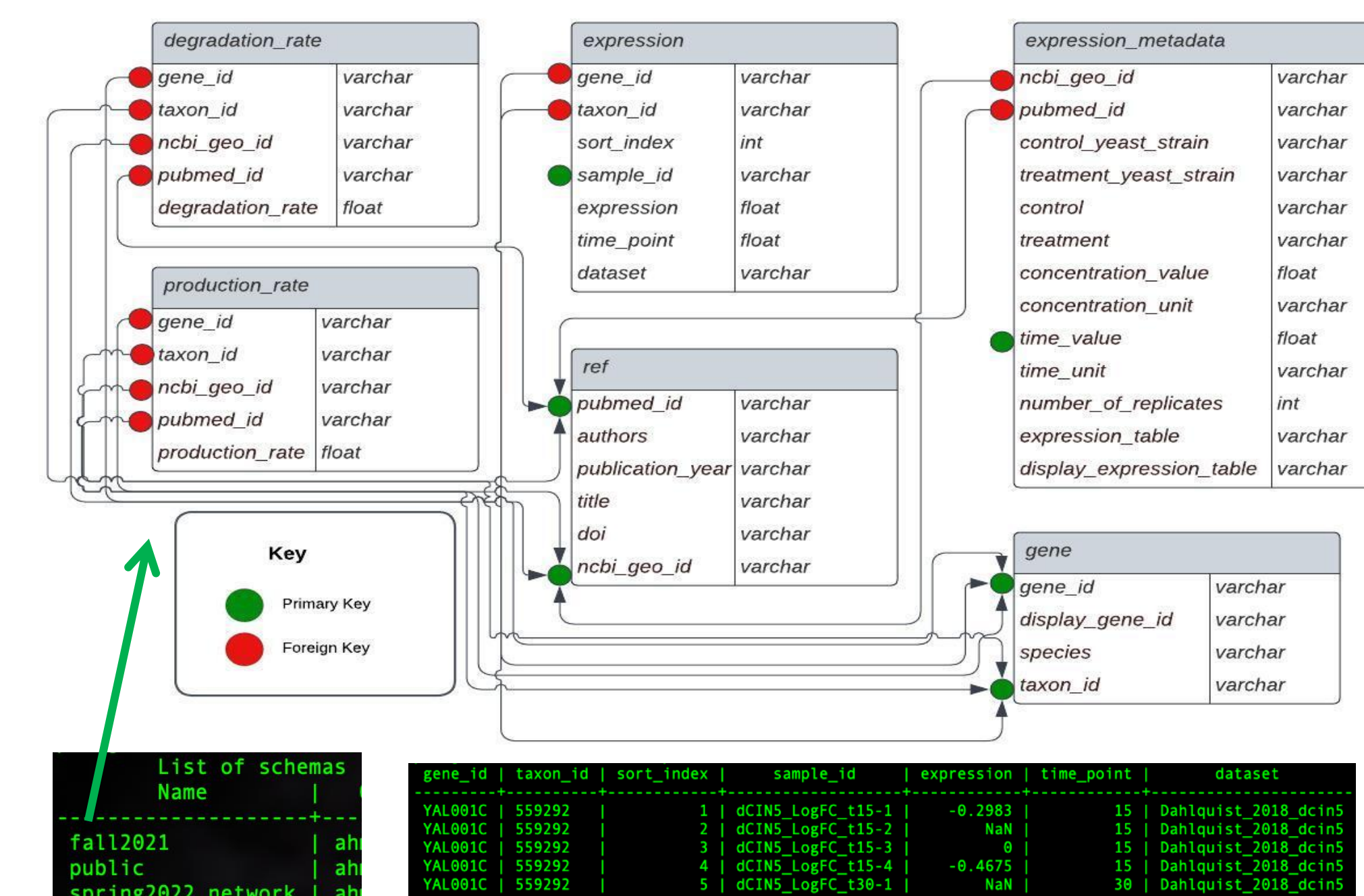
- 12-gene, 17 edge network from SGD
- Modeled with Kitagawa et al. (2002) expression data
- wt yeast treated with 75 μM thiamin for 15, 30, and 120 minutes



- 12-gene, 17 edge network from SGD
- Modeled with Thorsen et al. (2007) expression data
- wt yeast treated with 1 mM As(III) for 15, 30, and 60 minutes

Backend PostgreSQL Databases Store Gene Expression and Gene Regulatory Network Data

- Two backend PostgreSQL databases store expression and regulatory data that can be accessed by the user (see top, left code snippet).
- The expression database schema consists of 6 tables each containing information about genes and their corresponding expression data from previous experimental studies.
- For example, the Ref table lists the current datasets available for GRNsight to use.
- The default expression values for nodes in the network are based on expression data from Dahlquist_2018_wt dataset (see top right code snippet), but can be changed to other available options at any time.
- The users can also select genes to include in their network from the SGD data stored in GRNsight's regulation database (bottom code snippet).
- The Data Access Layer (DAL) module of the GRNsight database was refactored and expanded to reduce the amount of hard-coded information.
- Users can include yeast genes with irregular naming patterns in their network using the load from database function.
- When generating a network from the database, GRNsight correctly displays the node and edge count in the menu bar.
- The database was updated to store the timestamps in UTC instead of local timezones.
- The regulation database will be updated on a regular basis and users can retrieve data from all versions of the databases.
- Additional expression datasets will be added to the expression database in the future.

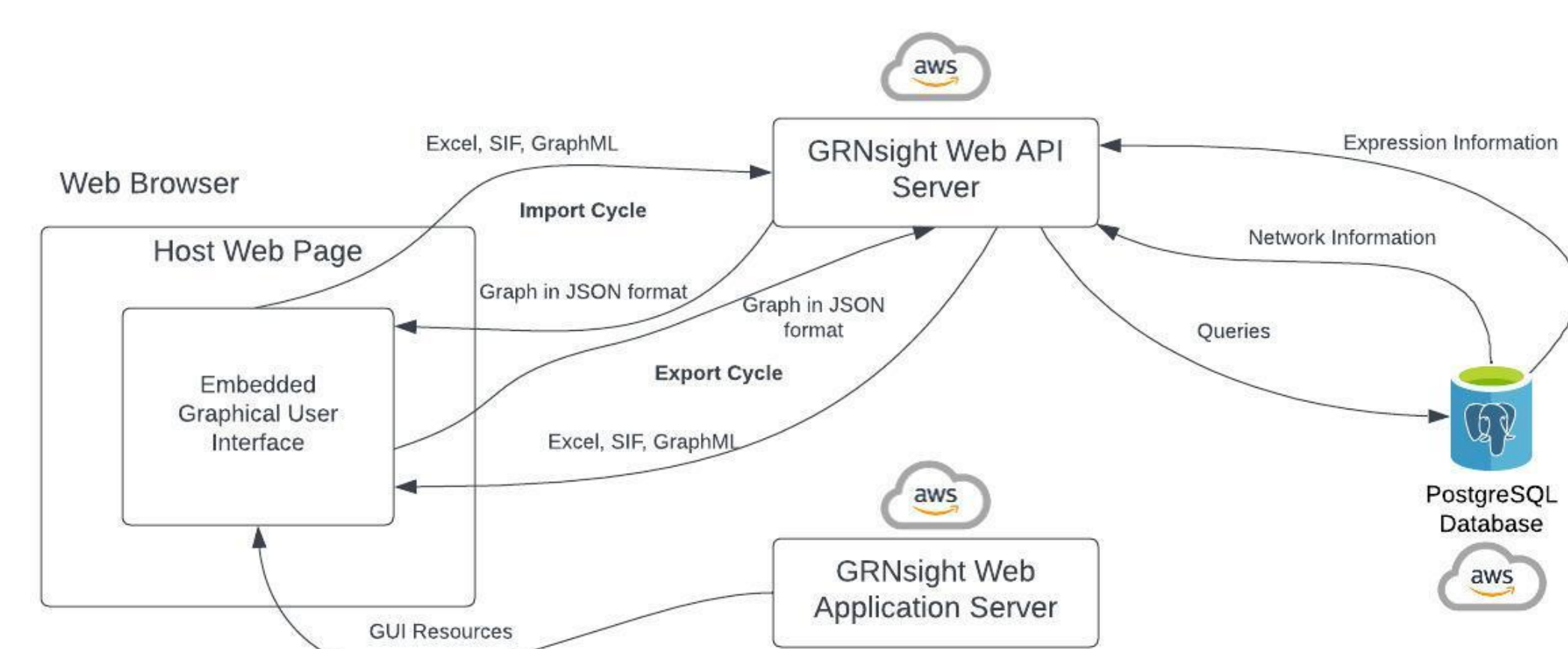


Information stored in expression table from expression database.

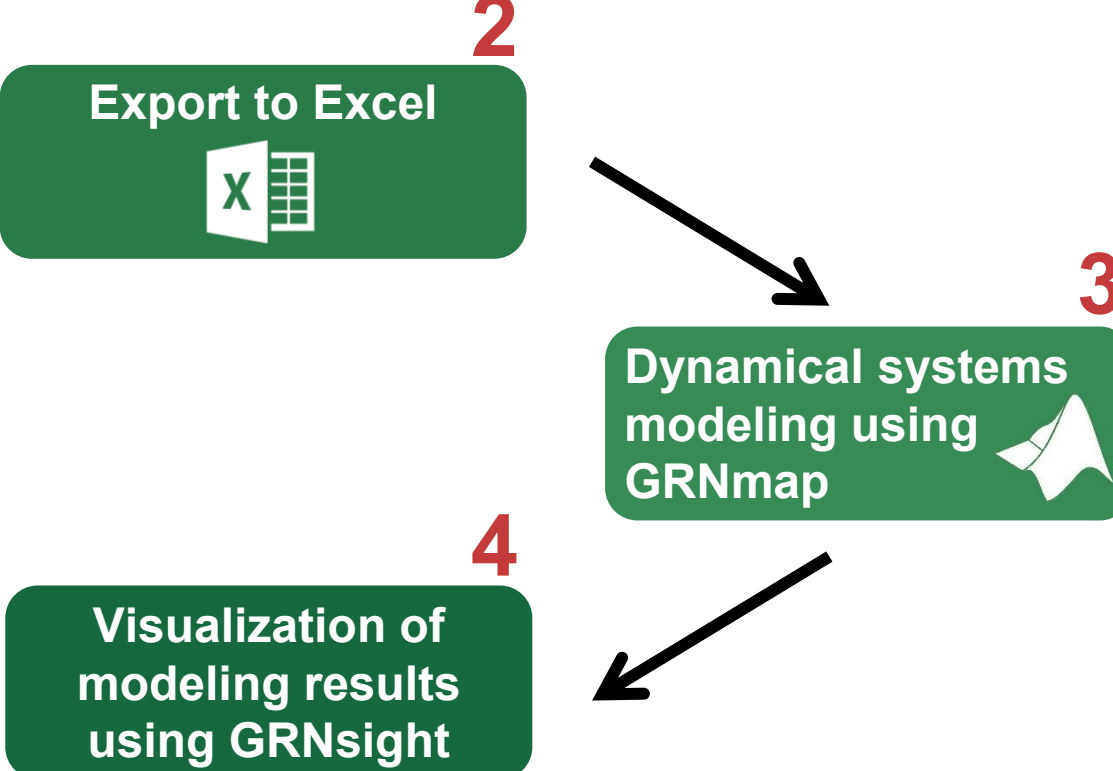
regulator_gene_id	target_gene_id	taxon_id	time_stamp	source
YAL001C	YAL002C	1	2002-09-23 18:17:52-07	YeastMine - Saccharomyces Genome Database
YAL001C	YAL002C	2	2002-09-23 18:17:52-07	YeastMine - Saccharomyces Genome Database
YAL001C	YAL002C	3	2002-09-23 18:17:52-07	YeastMine - Saccharomyces Genome Database
YAL001C	YAL002C	4	2002-09-23 18:17:52-07	YeastMine - Saccharomyces Genome Database
YAL001C	YAL002C	5	2002-09-23 18:17:52-07	YeastMine - Saccharomyces Genome Database

Information stored in network table from regulation database. Used to create networks.

GRNsight Architecture Facilitates Modeling-to-Visualization-to-Modeling Workflow



Workflow

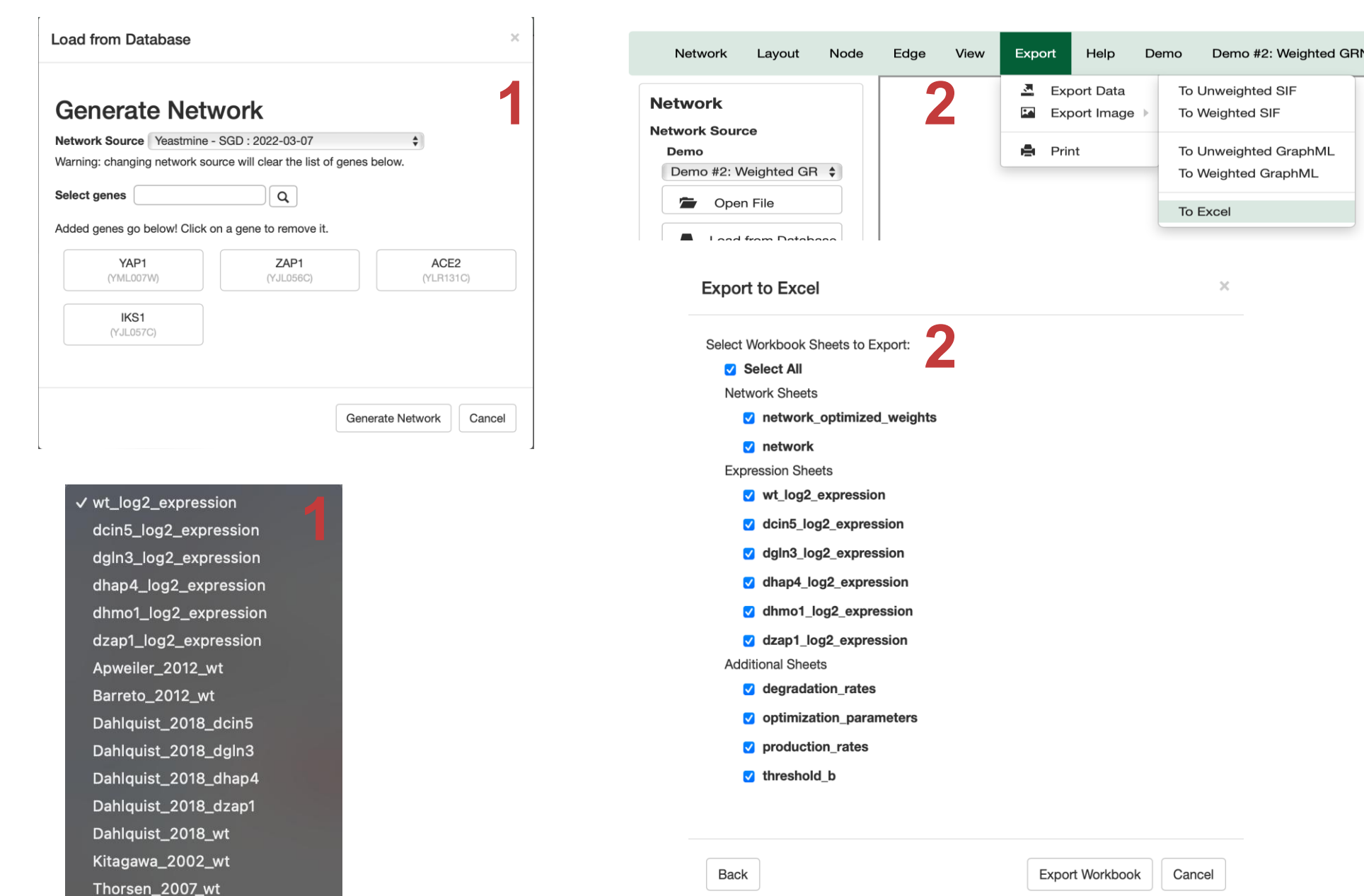


1. Selection of Network and Expression Data

- Users can select desired *Saccharomyces cerevisiae* (budding yeast) genes for their network using the data gathered from YeastMine.
- The users have the option to select expression data (node coloring) for their network from the current 5 datasets available in the GRNsight expression database.

2. Export to Excel

- Users then select which Excel workbook sheets they would like to export.
- The expression sheet options change based on the expression dataset selected.
- The Excel workbook generated is validated for input into the GRNmap MATLAB modeling software.



Future Directions

- To reduce the amount of hard-coded information in GRNsight code, a new GRNsettings table is being implemented in the database.
- Identify new bugs associated with current features.
- Work on outstanding bugs and feature enhancements to the graph display and user interface.
- Extend the backend database to store protein-protein interaction data.
- Allow undirected graphs so protein-protein interaction data can be displayed.

Availability

- GRNsight is free and open to all users and there is no login requirement.
- Web site content is available under the Creative Commons Attribution Non-Commercial Share Alike license.
- GRNsight code is available under the open-source BSD license.
- Usage is being tracked through Google Analytics.
- Current production software is at version 6.0.4.

New Users: 7,536

File Uploads: 7,836

Sessions: 18,470

Pageviews: 39,464

Acknowledgments

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- YeastRACT: <http://www.yeastRACT.com/index.php>
- SGD: <https://www.yeastgenome.org>
- GRNmap: <http://kdahlquist.github.io/GRNmap/>
- GRNsight: <https://dondi.github.io/GRNsight/>
- GRNsight's GitHub: <https://github.com/dondi/GRNsight>